

Relazione del laboratorio S6L5

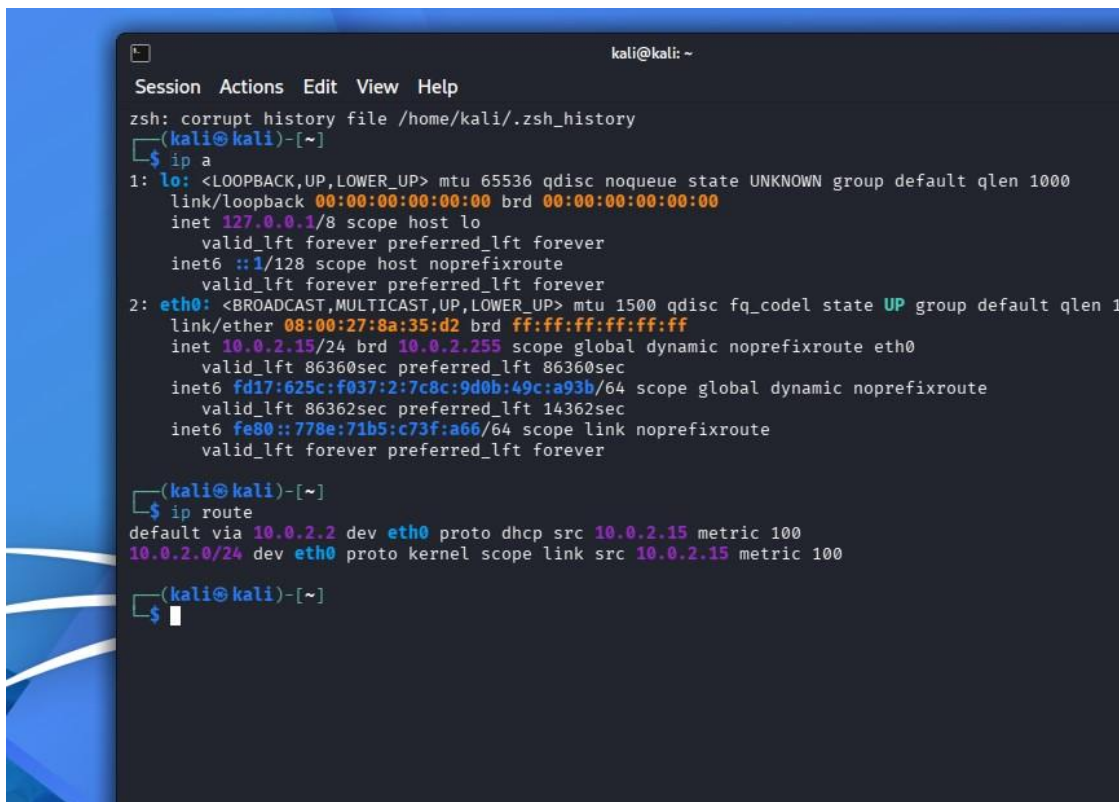
L'obiettivo di questa esercitazione è stato predisporre un ambiente di laboratorio dedicato allo studio della sicurezza informatica, configurando una macchina virtuale Kali Linux e installando i servizi necessari allo svolgimento delle attività previste dalla traccia.

Nel corso del laboratorio sono state eseguite le operazioni preliminari di configurazione della rete, l'installazione e la verifica del servizio OpenSSH, la creazione di un utente dedicato alle esercitazioni, l'installazione degli strumenti richiesti e la configurazione di un server FTP, verificandone il corretto funzionamento.

Successivamente è stato effettuato un test preliminare dell'ambiente, al solo scopo di confermare la corretta configurazione dei servizi e la piena operatività della macchina virtuale. L'analisi dettagliata degli strumenti impiegati e delle attività di verifica delle credenziali verrà approfondita in una relazione dedicata.

La presente relazione documenta quindi la preparazione dell'ambiente di lavoro, riportando le principali configurazioni eseguite e le relative evidenze raccolte durante il laboratorio mediante screenshot e brevi descrizioni tecniche.

1. Configurazione della rete



```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 86360sec preferred_lft 86360sec  
    inet6 fd17:625c:f037:2:7c8c:9d0b:49c:a93b/64 scope global dynamic noprefixroute  
        valid_lft 86362sec preferred_lft 14362sec  
    inet6 fe80::778e:71b5:c73f:a66/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
(kali@kali)-[~]  
$ ip route  
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100  
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100  
(kali@kali)-[~]  
$
```

Figura 1 – Verifica della configurazione di rete.

Output dei comandi ip a e ip route utilizzati per verificare l'indirizzo IP assegnato alla macchina Kali e la corretta configurazione del routing.

2. Configurazione del server FTP

```
secuzione] - Oracle VirtualBox
Aiuto

kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
└─$ sudo apt update && sudo apt install vsftpd -y
[sudo] password for kali:
Get:1 http://kali.mirror.garr.it/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.mirror.garr.it/kali kali-rolling/main amd64 Packages [21.4 MB]
Fetched 21.4 MB in 3s (6,148 kB/s)
1779 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  vsftpd

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1779
  Download size: 143 kB
  Space needed: 366 kB / 49.9 GB available

Get:1 http://kali.mirror.garr.it/kali kali-rolling/main amd64 vsftpd amd64 3.0.5-0.7 [143 kB]
Fetched 143 kB in 0s (313 kB/s)
Preconfiguring packages ...
Selecting previously unselected package vsftpd.
(Reading database... 432371 files and directories currently installed.)
Preparing to unpack ./vsftpd_3.0.5-0.7_amd64.deb...
Unpacking vsftpd (3.0.5-0.7)...
Setting up vsftpd (3.0.5-0.7)...
Creating group 'ftp' with GID 972.
Creating user 'ftp' (ftp daemon) with UID 972 and GID 972.
update-rc.d: We have no instructions for the vsftpd init script.
update-rc.d: It looks like a network service, we disable it.
vsftpd.service is a disabled or a static unit, not starting it.
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.1.5)...

(kali@kali)-[~]
└─$ sudo systemctl start vsftpd

(kali@kali)-[~]
└─$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-07-17 10:35:56 EDT; 10s ago
     Invocation: 01547857108044dabe952ebd65bffd7
       Docs: man:vsftpd(8)
    Process: 40173 ExecStartPre=/bin/mkdir -p /run/vsftpd/empty (code=exited, status=0/SUCCESS)
   Main PID: 40175 (vsftpd)
      Tasks: 1 (limit: 4456)
     Memory: 1.6M (peak: 2.5M)
        CPU: 18ms
    CGroup: /system.slice/vsftpd.service
            └─40175 /usr/sbin/vsftpd /etc/vsftpd.conf

Jul 17 10:35:56 kali systemd[1]: Starting vsftpd.service - vsftpd FTP server ...
Jul 17 10:35:56 kali systemd[1]: Started vsftpd.service - vsftpd FTP server.

(kali@kali)-[~]
└─$ sudo ss -tlnp | grep 21
LISTEN 0      32          *:21          *:21    users:(("vsftpd",pid=40175,fd=3))

(kali@kali)-[~]
└─$ ftp localhost
Trying [::1]:21 ...
Connected to localhost.
220 (vsFTPd 3.0.5)
Name (localhost:kali): test_user
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> bye
221 Goodbye.

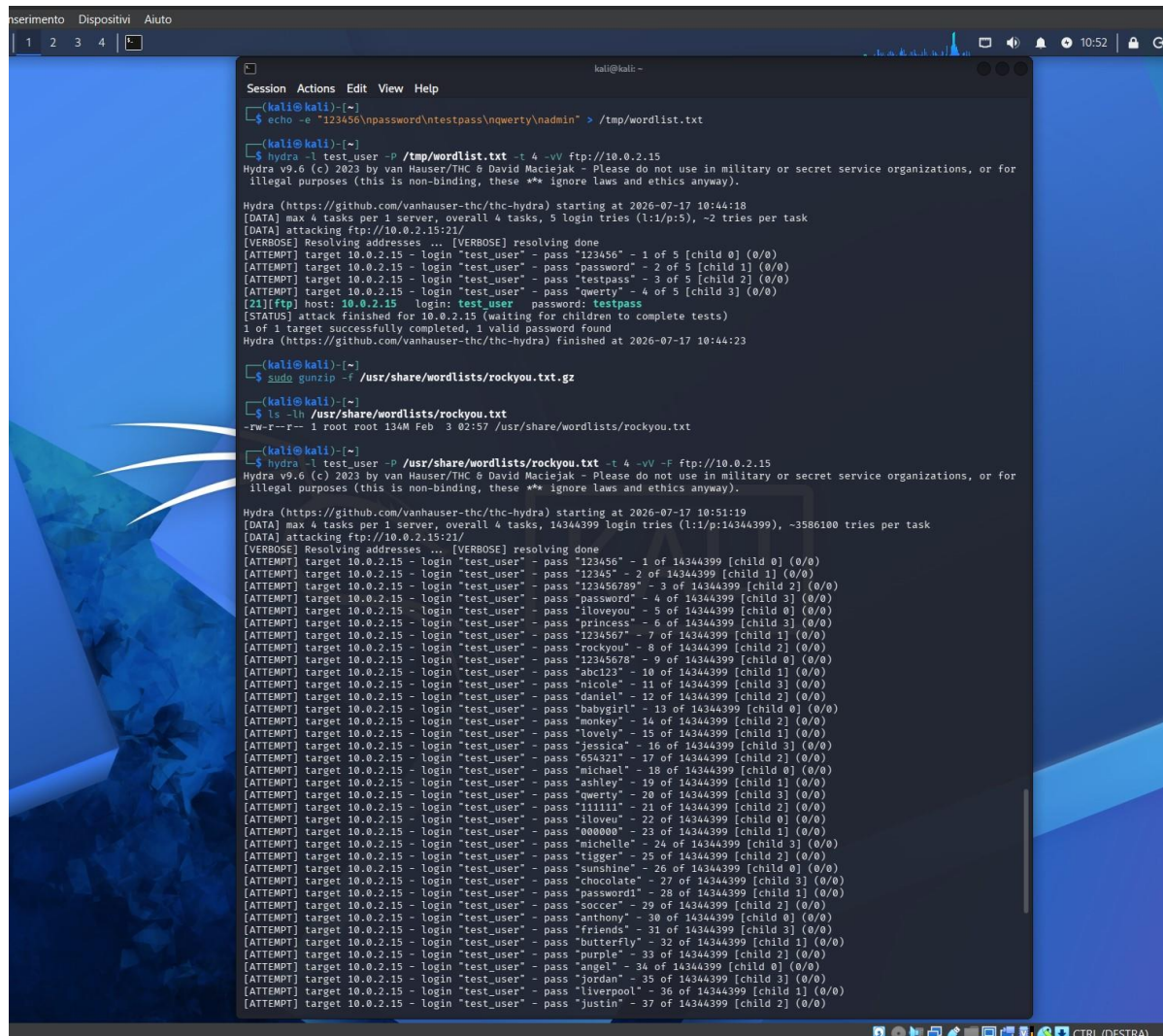
(kali@kali)-[~]
└─$ echo -e "123456\npassword\ntestpass\nqwerty\nadmin" > /tmp/wordlist.txt

(kali@kali)-[~]
```

Figura 2 – Installazione e verifica del servizio FTP.

Installazione di vsftpd, verifica dello stato del servizio, controllo della porta 21 e autenticazione FTP riuscita con l'utente di prova.

3. Preparazione dell'ambiente di test



```
kali@kali: ~  
$ echo -e "123456\npassword\ntestpass\nqwerty\nadmin" > /tmp/wordlist.txt  
$ hydra -l test_user -P /tmp/wordlist.txt -t 4 -vv ftp://10.0.2.15  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for  
illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-17 10:44:18  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 5 login tries (l:1/p:5), ~2 tries per task  
[DATA] attacking ftp://10.0.2.15:21/  
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "123456" - 1 of 5 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "password" - 2 of 5 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "testpass" - 3 of 5 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "qwerty" - 4 of 5 [child 3] (0/0)  
[21][ftp] host: 10.0.2.15 login: test_user password: testpass  
[STATUS] attack finished for 10.0.2.15 (waiting for children to complete tests)  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-17 10:44:23  
$ sudo gunzip -f /usr/share/wordlists/rockyou.txt.gz  
$ ls -lh /usr/share/wordlists/rockyou.txt  
-rw-r--r-- 1 root root 134M Feb 3 02:57 /usr/share/wordlists/rockyou.txt  
$ hydra -l test_user -P /usr/share/wordlists/rockyou.txt -t 4 -vv -F ftp://10.0.2.15  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for  
illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-17 10:51:19  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~3586100 tries per task  
[DATA] attacking ftp://10.0.2.15:21/  
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "123456" - 1 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "123457" - 2 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "123456789" - 3 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "password" - 4 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "iloveyou" - 5 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "princess" - 6 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "1234567" - 7 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "rockyou" - 8 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "12345678" - 9 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "abc123" - 10 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "nicole" - 11 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "daniel" - 12 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "babygirl" - 13 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "monkey" - 14 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "lovely" - 15 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "jessica" - 16 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "654321" - 17 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "michael" - 18 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "ashley" - 19 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "qwerty" - 20 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "111111" - 21 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "iloveu" - 22 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "000000" - 23 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "michelle" - 24 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "tiger" - 25 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "sunshine" - 26 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "chocolate" - 27 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "password1" - 28 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "soccer" - 29 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "anthony" - 30 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "friends" - 31 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "butterfly" - 32 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "purple" - 33 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "angel" - 34 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "Jordan" - 35 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "liverpool" - 36 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "justin" - 37 of 14344399 [child 2] (0/0)
```

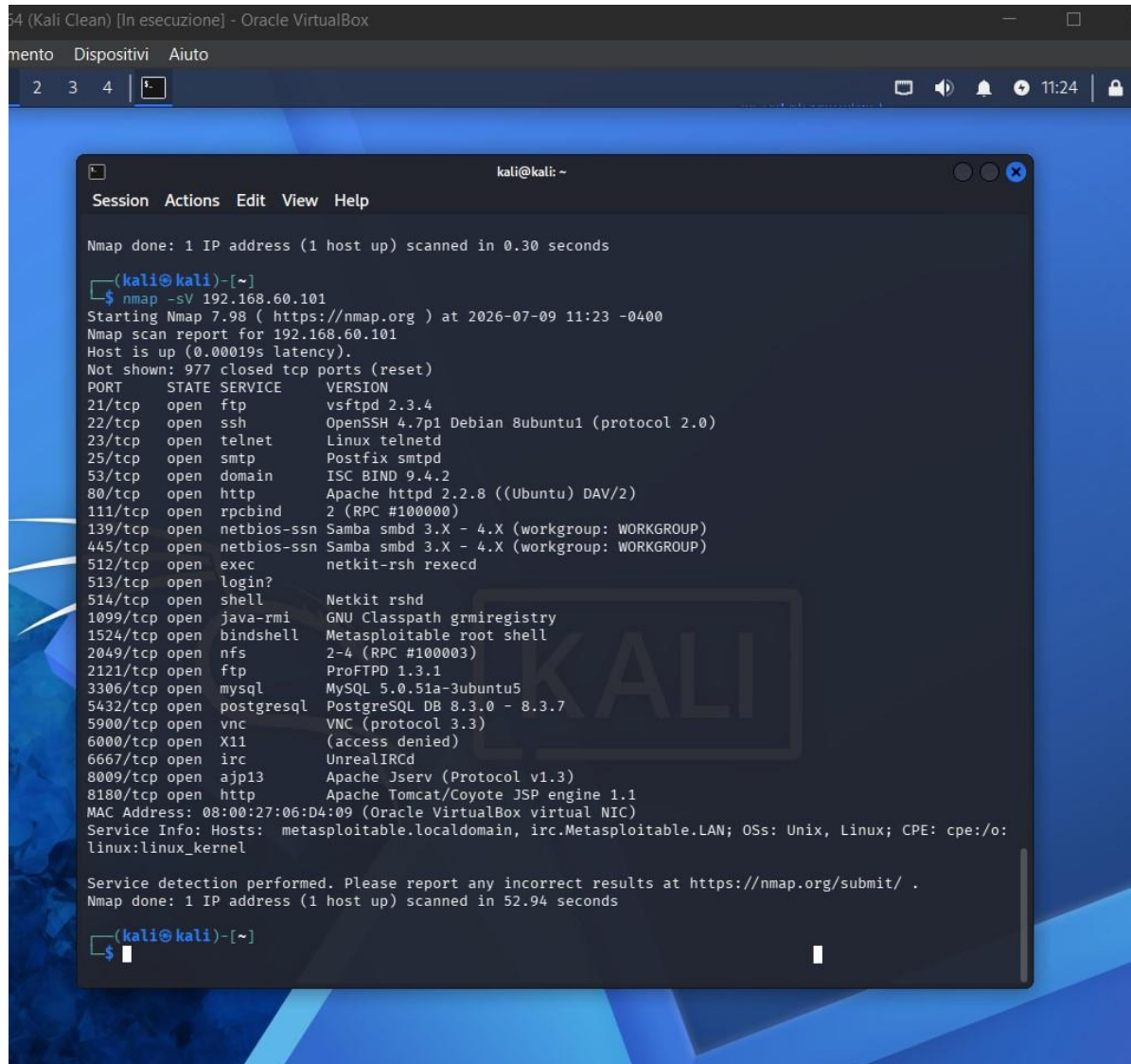
Figura 3 – Verifica preliminare dell'ambiente.

Breve verifica del corretto funzionamento dell'ambiente mediante Hydra. L'approfondimento dello strumento sarà trattato in una relazione dedicata.


```
kali@kali: ~  
Session Actions Edit View Help  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "5201314" - 415 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "jerome" - 416 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "gandako" - 417 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "muffin" - 418 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "gatita" - 419 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "babyko" - 420 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "246810" - 421 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "sweetheart" - 422 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "chivas" - 423 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "ladybug" - 424 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "kitty" - 425 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "popcorn" - 426 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "alberto" - 427 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "valeria" - 428 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "cookies" - 429 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "leslie" - 430 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "jenny" - 431 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "nicole1" - 432 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "12345678910" - 433 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "leonardo" - 434 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "jayjay" - 435 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "liliana" - 436 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "dexter" - 437 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "sexygirl" - 438 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "232323" - 439 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "amores" - 440 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "rockon" - 441 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "christ" - 442 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "babydoll" - 443 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "anthony1" - 444 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "marcus" - 445 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "bitch1" - 446 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "fatima" - 447 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "miamor" - 448 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "lover" - 449 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "christ1" - 450 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "single" - 451 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "eeyore" - 452 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "lalala" - 453 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "252525" - 454 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "scooter" - 455 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "natasha" - 456 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "skittles" - 457 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "brooklyn" - 458 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "colombia" - 459 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "159357" - 460 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "teddybear" - 461 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "winnie" - 462 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "happy" - 463 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "manutd" - 464 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "123456a" - 465 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "britney" - 466 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "katrina" - 467 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "christina" - 468 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "pasaway" - 469 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "cocacola" - 470 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "mahal" - 471 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "grace" - 472 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "linda" - 473 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "albert" - 474 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "tatiana" - 475 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "london" - 476 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "cantik" - 477 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "0123456" - 478 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "lakers" - 479 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "marie" - 480 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "teiubesc" - 481 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "147258369" - 482 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "charlotte" - 483 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "natalia" - 484 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "francisco" - 485 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "amorcito" - 486 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.2.15 - login "test_user" - pass "smile" - 487 of 14344399 [child 1] (0/0)
```

Figura 4 – Prova con wordlist estesa.
Esempio di esecuzione con una wordlist di grandi dimensioni per osservare il comportamento dell'ambiente durante una verifica prolungata. Ho calcolato che ci sarebbero volute 16 ore circa, quindi ho interrotto dopo qualche minuto.

4. Individuazione dei servizi della macchina di laboratorio



```
kali@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds  
(kali@kali)-[~]  
$ nmap -sV 192.168.60.101  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-09 11:23 -0400  
Nmap scan report for 192.168.60.101  
Host is up (0.00019s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE        VERSION  
21/tcp    open  ftp            vsftpd 2.3.4  
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet         Linux telnetd  
25/tcp    open  smtp           Postfix smtpd  
53/tcp    open  domain         ISC BIND 9.4.2  
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind        2 (RPC #100000)  
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec           netkit-rsh rexecd  
513/tcp   open  login?           
514/tcp   open  shell          Netkit rshd  
1099/tcp  open  java-rmi       GNU Classpath grmiregistry  
1524/tcp  open  bindshell      Metasploitable root shell  
2049/tcp  open  nfs            2-4 (RPC #100003)  
2121/tcp  open  ftp            ProFTPD 1.3.1  
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc            VNC (protocol 3.3)  
6000/tcp  open  X11            (access denied)  
6667/tcp  open  irc            UnrealIRCd  
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)  
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1  
MAC Address: 08:00:27:06:D4:09 (Oracle VirtualBox virtual NIC)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 52.94 seconds  
(kali@kali)-[~]  
$
```

Figura 5 – Identificazione di porte e servizi.
Scansione Nmap per identificare i servizi disponibili sulla macchina di laboratorio.

```
(kali@kali)-[~]
$ nmap -A 192.168.60.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-09 11:30 -0400
Nmap scan report for 192.168.60.101
Host is up (0.00036s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_Connected to 192.168.60.100
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_vsFTPD 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_smtp-command: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDST
ATUSCODES, 8BITMIME, DSN
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|_program version port/proto service
|_100000 2 111/tcp rpcbind
|_100000 2 111/udp rpcbind
|_100003 2,3,4 2049/tcp nfs
|_100003 2,3,4 2049/udp nfs
|_100005 1,2,3 52437/udp mountd
|_100005 1,2,3 55662/tcp mountd
|_100021 1,3,4 43912/udp nlockmgr
|_100021 1,3,4 44236/tcp nlockmgr
|_100024 1 37437/tcp status
|_100024 1 51663/udp status
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
|_mysql-info:
|_Protocol: 10
|_Version: 5.0.51a-3ubuntu5
|_Thread ID: 9
|_Capabilities flags: 43564
|_Some Capabilities: LongColumnFlag, Support41Auth, ConnectWithDatabase, SupportsTransactions, Support
sCompression, SwitchToSSLAfterHandshake, Speaks41ProtocolNew
|_Status: Autocommit
|_Salt: @n+*.JgJ|Rc{c[#6'`1q
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
```

Figura 6 – Scansione avanzata.

Avvio di una scansione avanzata finalizzata alla raccolta di informazioni dettagliate sui servizi esposti.

Conclusioni parziali

Le verifiche effettuate confermano il corretto funzionamento dell'ambiente di laboratorio. Tutti i servizi risultano operativi e l'infrastruttura è pronta per completare l'esercitazione. L'analisi dettagliata di Hydra sarà sviluppata in un documento separato.